

Name:

Student ID:

OPTIMISATION PROBLEM SHEET

This sheet covers the three optimisation lectures. Question 1 looks at gradient descent and Newton's method on simple functions, where we can see exactly what happens. Questions 2 and 3 use Lagrange multipliers and duality. Question 4 asks you to check some of these results in Python. The early parts of each question are short; the later parts ask you to put the pieces together.

1 Gradient descent with learning rate r updates $w \leftarrow w - rf'(w)$ (or $w \leftarrow w - r\nabla f(w)$ in several dimensions). In this question we see exactly when it works on quadratic functions.

- (a) For $f(w) = \frac{1}{2}aw^2$ with $a > 0$, write down the gradient descent update and show that after t steps $w_t = (1 - ra)^t w_0$. [3 marks]

3

- (b) For which learning rates does $w_t \rightarrow 0$ (the minimum)? Which learning rate is best? Describe what happens when $r = 1.5/a$ and when $r = 2.5/a$. [4 marks]

4

- (c) Now take $f(\mathbf{w}) = \frac{1}{2}(aw_1^2 + bw_2^2)$ with $a \geq b > 0$. Show that the two components evolve independently, and find the condition on r for convergence. Show that the fixed learning rate that makes the slower of the two components converge fastest is $r = 2/(a + b)$, and that the error is then multiplied by $(\kappa - 1)/(\kappa + 1)$ each step, where $\kappa = a/b$ is the condition number. [5 marks]

5

- (d) Take $a = 100$ and $b = 1$. What is the largest stable learning rate, and the best one? Roughly how many gradient descent steps are needed to reduce the error by a factor of 10^6 ? How many steps would Newton's method need? [4 marks]

4

- (e) Newton's method in one dimension updates $w \leftarrow w - f'(w)/f''(w)$. For $f(w) = e^w - 2w$, show that the minimum is at $w^* = \ln 2$ and that Newton's update is $w \leftarrow w - 1 + 2e^{-w}$. Starting from $w_0 = 0$, use a calculator to compute four iterations and the error $|w_t - \ln 2|$. What happens to the error? [4 marks]

4

End of question 1

(a) $\frac{3}{3}$	(b) $\frac{4}{4}$	(c) $\frac{5}{5}$	(d) $\frac{4}{4}$	(e) $\frac{4}{4}$	Total $\frac{20}{20}$
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2 To minimise (or maximise) $f(x)$ subject to $g(x) = 0$ we form the Lagrangian $\mathcal{L}(x, \alpha) = f(x) - \alpha g(x)$ and set its derivatives with respect to x and α to zero. For an inequality constraint $g(x) \geq 0$ the KKT conditions add $\alpha \geq 0$ and $\alpha g(x) = 0$.

- (a) A rectangle has sides x and y and a fixed perimeter $2x + 2y = P$. Use a Lagrange multiplier to find the sides that maximise its area $A = xy$. Show that α equals dA^*/dP , where A^* is the maximum area: the Lagrange multiplier tells us how much the optimum improves if we relax the constraint. [5 marks]

5

- (b) Find the point x closest to the origin on the hyperplane $a^T x = b$, by minimising $\|x\|^2$ subject to this constraint. Hence show that the distance from the origin to the hyperplane is $|b|/\|a\|$. (This is the geometry behind the margin of a support vector machine.) [6 marks]

6

- (c) Minimise $f(x,y) = (x-2)^2 + (y-1)^2$ subject to the inequality $x + y \leq 2$. Write the constraint in the form $g(x,y) \geq 0$, decide whether it is active, and find the solution and the value of α . [3 marks]

$\frac{3}{3}$

- (d) Repeat with the constraint $x + y \leq 4$. [3 marks]

$\frac{3}{3}$

- (e) Finally, take the *equality* constraint $x + y = 4$. Find the solution and α . Why would a negative α be impossible for the inequality constraint $x + y \leq 4$? [3 marks]

$\frac{3}{3}$

End of question 2

(a) $\frac{5}{5}$ (b) $\frac{6}{6}$ (c) $\frac{3}{3}$ (d) $\frac{3}{3}$ (e) $\frac{3}{3}$ Total $\frac{20}{20}$

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3 In the lectures we saw that the linear programme (the *primal*)

$$\min_x c^T x \quad \text{subject to} \quad Mx \geq b, \quad x \geq 0$$

has the *dual*

$$\max_{\alpha} b^T \alpha \quad \text{subject to} \quad M^T \alpha \leq c, \quad \alpha \geq 0.$$

- (a) Solve the linear programme: minimise $2x_1 + 3x_2$ subject to $x_1 + x_2 \geq 4$, $x_1 + 3x_2 \geq 6$ and $x_1, x_2 \geq 0$. (The optimum of a linear programme is at a corner of the feasible region, so check the corners.) Write down the dual, solve it in the same way, and compare the two optimal values. [6 marks]

6

- (b) Show that for *any* x satisfying the primal constraints and *any* α satisfying the dual constraints, $b^T \alpha \leq c^T x$. Why is this useful? [4 marks]

4

- (c) If the primal has n variables and k constraints (not counting $x \geq 0$), how many variables and constraints does the dual have? Which would you rather solve when $n = 10^6$ and $k = 10$? [3 marks]

16

- (d) Now consider the quadratic programme: minimise $x^T x$ subject to $\mathbf{M}x = b$, where $\mathbf{M}$ is $k \times n$ with $k < n$. Form the Lagrangian, find $x^*(\alpha)$, and show that the dual problem is

$$\max_{\alpha} -\frac{1}{4}\alpha^T \mathbf{M} \mathbf{M}^T \alpha + \alpha^T b.$$

Solve it, and show that $\mathbf{x}^* = \mathbf{M}^T(\mathbf{M}\mathbf{M}^T)^{-1}\mathbf{b}$. How big is the matrix we have to invert? Check that $k = 1$ recovers question 2(b). [7 marks]

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End of question 3

(a) $\frac{6}{6}$ (b) $\frac{4}{4}$ (c) $\frac{3}{3}$ (d) $\frac{7}{7}$ Total $\frac{20}{20}$

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4 In this question you check some of these results in Python (or any other language). The following may be useful.

<code>import numpy as np</code>	import the numpy module
<code>H = np.diag([100.0, 1.0])</code>	a diagonal matrix
<code>np.array([1.0, 1.0])</code>	a vector
<code>H @ w</code>	matrix–vector multiplication
<code>np.linalg.norm(w)</code>	the length $\ w\ $
<code>rng = np.random.default_rng(1)</code>	a random number generator with a fixed seed, so results can be repeated
<code>rng.integers(0, m, size=10)</code>	10 random indices, for a mini-batch

- (a) Run gradient descent on $f(w) = \frac{1}{2}(100w_1^2 + w_2^2)$ from $w = (1,1)^\top$. For $r = 0.005$, $r = 2/101$ and $r = 0.0201$, count the steps until $\|w\| < 10^{-6}$ (or report that it diverges). Compare with question 1. [5 marks]

5

- (b) Add momentum, $v \leftarrow \gamma v - r \nabla f(w)$, $w \leftarrow w + v$. Count the steps for $\gamma = 0.9$, $r = 0.01$, and for the tuned values $r = 4/121$, $\gamma = (9/11)^2$. Compare with plain gradient descent. [5 marks]

5

